

Operators and Logic

Where We Left Off

A **variable** is one name for one value

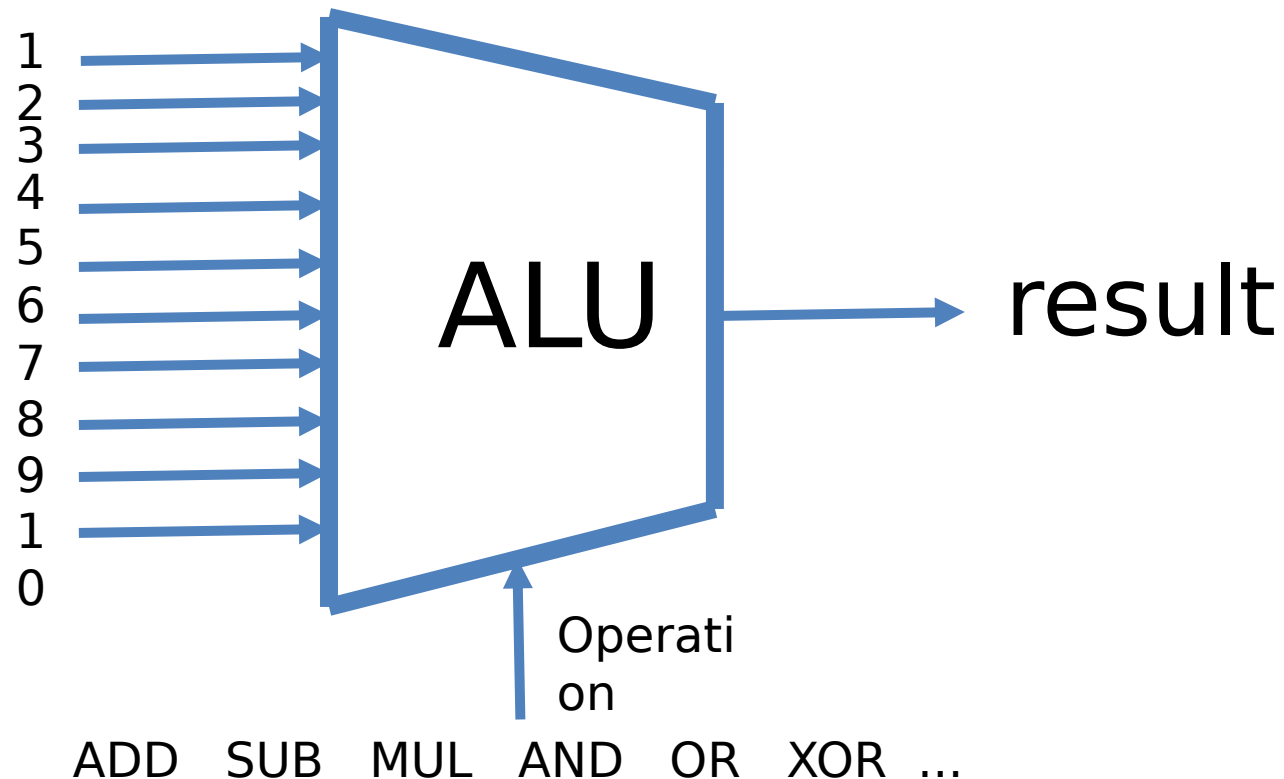
A **struct** is a new type $\langle T \rangle$ built from members

An **array** is N instances of one type

Now: what can we **do** with them?

The ALU Does the Work

The ALU from lecture one did an **ADD**
The **Operation** input selects what it does



What an Operator Is

An **operator** takes the contents of its **operands** and produces a new value

result $\leftarrow (a) + (b)$

Binary operators take two operands: $a + b$

Unary operators take one: $-a$ $!a$

Arithmetic Operators

+	addition
-	subtraction
*	multiplication
/	division
%	remainder (modulo)

The **same symbols** in C, C++, and Rust

Integer Division Truncates

$$7 / 2 = 3$$

$$-7 / 2 = -3$$

$$7.0 / 2.0 = 3.5$$

Integer division throws away the fraction

It rounds **toward zero** in all three languages

The Remainder Operator

$7 \% 3 = 1$

$-7 \% 3 = -1$ sign follows the left operand

C, C++: $\%$ is for **integers only**
use `fmod(a, b)` for floating point

Rust: $\%$ also works on `f32` and `f64`

Division by Zero

Integers:

C, C++: **undefined behavior**

Rust: the program **panics** and stops

Floating point, all three languages:

$1.0 / 0.0 = \text{inf}$

$0.0 / 0.0 = \text{NaN}$ (not a number)

Increment and Decrement

C, C++:

`x++ ;` add one

`x-- ;` subtract one

Rust has **no ++ or -**

You will need to use the compound operators

`x += 1 ;`

`x -= 1 ;`

Prefix and Postfix

`x = 5 ; y = x++ ;` `y` is 5, `x` is 6

`x = 5 ; y = ++x ;` `y` is 6, `x` is 6

Postfix: use the value, **then** add one

Prefix: add one, **then** use the value

Never use `x++` twice in one expression

Compound Assignment

$x += 5 ;$	$x \leq (x) + 5$
$x -= 5 ;$	$x \leq (x) - 5$
$x *= 5 ;$	$x \leq (x) * 5$
$x /= 5 ;$	$x \leq (x) / 5$
$x \% = 5 ;$	$x \leq (x) \% 5$

Same in C, C++, and Rust

Overflow in C and C++

```
uint8_t x = 255 ;  
x = x + 1 ;      x is now 0
```

Unsigned overflow wraps around - defined

Signed overflow is **undefined behavior**

```
int8_t y = 127 ; y + 1  
-> ???
```

Overflow in Rust

```
let x : u8 = 255 ;  
let y = x + 1 ;
```

Debug build:	panics - overflow is caught
Release build:	wraps to 0

Say what you mean:

```
x.wrapping_add(1) 0  
x.checked_add(1)  None  
x.saturating_add(1) 255
```

Mixing Types in C and C++

```
uint8_t  a = 200, b = 100 ;
```

```
int      d = a + b ;      300
```

```
uint8_t  c = a + b ;      44
```

Small types are **promoted** to int first
The result is cut down when stored

Mixing Types in Rust

```
let a : u8 = 200 ;  
let d : i32 = a + 1000 ;      error
```

Rust **never** converts types for you

You convert explicitly with **as**:

```
let d : i32 = a as i32 + 1000 ;
```

Floating Point is Not Exact

`0.1 + 0.2 == 0.3` **false**

Most decimals cannot be stored exactly in binary

Compare with a tolerance instead:

`fabs(a - b) < 1e-6` C, C++

`(a - b).abs() < 1e-6` Rust

Comparison Operators

`==` equal

`!=` not equal

`<` less than

`>` greater than

`<=` less or equal

`>=` greater or equal

The result is true or false

C: an int, 1 or 0

C++, Rust: a bool

= is not ==

if (x = 5) assigns 5, then tests it
always true in C and C++

if (x == 5) tests whether x is 5

Rust refuses to compile the first one
an assignment is not a bool

Chained Comparisons

Mathematics: $0 < x < 10$

C, C++: compiles, and is **wrong**
($0 < x$) is 0 or 1 - always < 10

Rust: compile error

Write: $0 < x \ \&\& \ x < 10$

Operators on Struct Members

A member is an ordinary variable
Every operator works on it:

```
my_sample.psi * 2.0  
my_sample.adc_counts + 1  
my_sample.timestamp_ms - start
```

Same in C, C++, and Rust

Assigning a Whole Struct

C, C++:

```
sample_t sample2 ;  
sample2 = my_sample ;    copies every byte
```

Rust:

```
let sample2 = my_sample ;    moves it  
my_sample can no longer be used
```

Add `#[derive(Clone, Copy)]` to copy instead

Comparing Structs in C

if (my_sample == sample2) **compile error**

Compare member by member:

```
my_sample.adc_counts == sample2.adc_counts  
&& my_sample.psi == sample2.psi  
&& my_sample.timestamp_ms ==  
sample2.timestamp_ms
```

Do not use memcmp - it compares the **padding** too

Comparing Structs in C++

Ask the compiler to write == for you (C++20):

```
struct sample {  
    uint32_t    adc_counts ;  
    float      psi ;  
    uint64_t    timestamp_ms ;  
    bool operator==(const sample&) const = default ;  
} ;
```

Now `my_sample == sample2` compares every member

Comparing Structs in Rust

```
#[derive(PartialEq)]  
struct Sample {  
    adc_counts : u32,  
    psi : f32,  
    timestamp_ms : u64,  
}
```

Now `my_sample == sample2` compares every member

Arithmetic on Whole Structs

`my_sample + sample2 ??`

No language can guess what `+` means for your type

C: not possible - write a function

C++: define `operator+`

Rust: implement the `Add` trait

This is **operator overloading** - later lecture

Operators on Array Elements

An element is an ordinary variable
Every operator works on it:

`counts[i] + 1`

`psi[i] * scale`

`history[7].psi - history[6].psi`

Same in C, C++, and Rust

Whole Arrays in C

```
uint32_t a[4], b[4] ;
```

```
a = b ;
```

compile error

```
if ( a == b )
```

compares **addresses**, not
contents

Copy with: `memcpy(a, b, sizeof(a)) ;`

Compare with: a loop, or `memcmp`

Whole Arrays in C++

```
#include <array>
```

```
std::array<uint32_t, 4> a, b ;
```

```
a = b ;           copies every element
```

```
if ( a == b )     compares every element
```

Another reason to prefer std::array

Whole Arrays in Rust

```
let a : [u32 ; 4] = [1, 2, 3, 4] ;
```

```
let mut b : [u32 ; 4] = [0 ; 4] ;
```

```
b = a ;
```

copies every element

```
if a == b { ... }
```

compares every element

Works because the length is part of the type

Element-wise Math Needs a Loop

C, C++:

```
for ( int i = 0 ; i < 4 ; i++ ) {  
    sum[i] = a[i] + b[i] ;  
}
```

Rust:

```
for i in 0..4 {  
    sum[i] = a[i] + b[i] ;  
}
```

Two Kinds of Logic

Clause level

combines conditions that are true or false
if (psi < 10.0 || psi > 80.0)

Bit level

applies logic to **every bit** of an integer
status & 0x80

What Counts as True

C: 0 is false, **anything else** is true
if (count) means if (count != 0)

C++: bool, but numbers convert silently

Rust: only a **bool** is a condition
if count { } **compile error**
if count != 0 { } correct

Logical Operators

&&	AND	true if both are true
	OR	true if either is true
!	NOT	flips true and false

The **same symbols** in C, C++, and Rust

```
if ( psi < 10.0 || psi > 80.0 ) alarm() ;
```

Short-Circuit Evaluation

`a && b` if `a` is false, `b` is **never evaluated**

`a || b` if `a` is true, `b` is **never evaluated**

This makes guards safe:

`if ( count != 0 && total / count > 10 )`

Division by zero cannot happen

Boolean Logic

George Boole, 1854: logic as **algebra**

Two values: 0 and 1 (false and true)

Every operation is defined completely by a **truth table**: one row for every input combination

Two inputs $\rightarrow$ 4 rows

NOT

$$Y = A'$$

Flips the input

Clause level:

`!a`

all three

Bit level:

`~a`

C, C++

`!a`

Rust

A	Y
0	1
1	0

AND

$$Y = A \cdot B$$

1 only when **both** inputs are 1

Clause level:

a && b

all three

Bit level:

a & b

all three

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR

$$Y = A + B$$

1 when **either** input is 1
(or both)

Clause level:

$a \vee b$

all three

Bit level:

$a \mid b$

all three

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NAND

$$Y = (A \cdot B)'$$

NOT of AND: 0 only when
both inputs are 1

Clause level:

!(a && b) all three

Bit level:

~(a & b) C, C++

!(a & b) Rust

A	B	Y
0	0	1
0	1	1
1	0	1
1	1	0

NOR

$$Y = (A + B)'$$

NOT of OR: 1 only when
both inputs are 0

Clause level:

$!(a \parallel b)$

all three

Bit level:

$\sim(a \mid b)$

C, C++

$!(a \mid b)$

Rust

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	0

XOR

$$Y = A \oplus B$$

1 when the inputs **differ**

Clause level:

$a \neq b$

$a \wedge b$

Bit level:

$a \wedge b$

C, C++

Rust (bool)

all three

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	0

XNOR

$$Y = (A \oplus B)'$$

1 when the inputs are **equal**

Clause level:

`a == b`

all three

Bit level:

`~(a ^ b)`

`!(a ^ b)`

C, C++

Rust

A	B	Y
0	0	1
0	1	0
1	0	0
1	1	1

All Seven Together

A	B	NOT A	AND	OR	NAND	NOR	XOR	XNOR
0	0	1	0	0	1	1	0	1
0	1	1	0	1	1	0	1	0
1	0	0	0	1	1	0	1	0
1	1	0	1	1	0	0	0	1

NOT uses only A. The shaded columns are outputs.

NAND Can Build Everything

NOT A = A NAND A

A AND B = NOT (A NAND B)

A OR B = (NOT A) NAND (NOT B)

NAND is **universal**

A whole computer can be built from one kind of gate

De Morgan's Laws

$\text{NOT} (A \text{ AND } B) = (\text{NOT } A) \text{ OR } (\text{NOT } B)$

$\text{NOT} (A \text{ OR } B) = (\text{NOT } A) \text{ AND } (\text{NOT } B)$

In code:

`!(a && b) == !a || !b`

`!(a || b) == !a && !b`

Flip every operator and negate every term

Bitwise Operators

&	AND	all three
	OR	all three
^	XOR	all three
~	NOT	C, C++
!	NOT	Rust
<<	shift left	all three
>>	shift right	all three

Each bit column is computed **independently**

Bitwise AND

Each column is one row of the AND truth table
Result bit is 1 only where **both** bits are 1

[illegible]

Bitwise OR

Each column is one row of the OR truth table
Result bit is 1 where **either** bit is 1

	7	6	5	4	3	2	1	0	
a	1	1	0	0	1	0	1	0	0xCA = 202
b	1	1	1	1	0	0	0	0	0xF0 = 240
a b	1	1	1	1	1	0	1	0	0xFA = 250

Bitwise XOR

Each column is one row of the XOR truth table
Result bit is 1 where the bits **differ**

	7	6	5	4	3	2	1	0	
a	1	1	0	0	1	0	1	0	0xCA = 202
b	1	1	1	1	0	0	0	0	0xF0 = 240
a ^ b	0	0	1	1	1	0	1	0	0x3A = 58

Bitwise NOT

Every bit flips

$\sim a$ in C and C++, $!a$ in Rust

	<i>7</i>	<i>6</i>	<i>5</i>	<i>4</i>	<i>3</i>	<i>2</i>	<i>1</i>	<i>0</i>	
<i>a</i>	1	1	0	0	1	0	1	0	$0xCA = 202$
$\sim a$	0	0	1	1	0	1	0	1	$0x35 = 53$

& is not &&

C: 1 & 2 = 0 bitwise: 01 & 10 = 00
 1 && 2 = 1 both are non-zero

Rust: 1 & 2 = 0
 1 && 2 **compile error** - not bool

if (a & b) is a different test than if (a && b)

Shift Left

Every bit moves one place left; a 0 enters on the right

$x \ll n$ multiplies by 2 to the n

	7	6	5	4	3	2	1	0	
a	1	1	0	0	1	0	1	0	202
a << 1	1	0	0	1	0	1	0	0	148 (bit 7 fell off)

Shift Right

Every bit moves one place right; bit 0 falls off
 $x \gg n$ divides by 2 to the n

	7	6	5	4	3	2	1	0	
a	1	1	0	0	1	0	1	0	202
a >> 1	0	1	1	0	0	1	0	1	101

Shifting Signed Values

Unsigned: a **0** enters on the left (logical)

Signed:
(arithmetic) the **sign bit** is copied

$$-8 \gg 1 = -4$$

Rust, C++20: guaranteed arithmetic
C: implementation defined

Use unsigned types for bit work

Shifting Too Far

```
uint32_t x = 1 ;  
x << 32      ??
```

C, C++: **undefined behavior**

Rust: panics in a debug build

Shift by less than the width of the type

Use `1u << n` in C - never `1 << 31`

Bitwise Compound Assignment

$x \&= m ;$	$x \leqslant = (x) \& m$
$x = m ;$	$x \leqslant = (x) m$
$x \wedge = m ;$	$x \leqslant = (x) \wedge m$
$x \ll = n ;$	$x \leqslant = (x) \ll n$
$x \gg = n ;$	$x \leqslant = (x) \gg n$

Same in C, C++, and Rust

Masks

A **mask** is a constant that selects bits

C: `#define ENABLE_BIT ( 1u << 7 )`

C++: `constexpr uint8_t ENABLE_BIT = 1u << 7 ;`

Rust: `const ENABLE_BIT : u8 = 1 << 7 ;`

Constants are **SCREAMING_SNAKE_CASE**
in all three languages

Set a Bit

$x \mid= (1 \ll 3);$ OR forces bit 3 to 1
Every other bit is unchanged

	7	6	5	4	3	2	1	0	
x	0	1	0	0	0	0	0	1	
$1 \ll 3$	0	0	0	0	1	0	0	0	mask
x mask	0	1	0	0	1	0	0	1	bit 3 set

Clear a Bit

`x &= ~( 1u << 6 );` C, C++
`x &= !( 1 << 6 );` Rust

	7	6	5	4	3	2	1	0	
x	0	1	0	0	1	0	0	1	
$\sim(1 \ll 6)$	1	0	1	1	1	1	1	1	inverted mask
x & mask	0	0	0	0	1	0	0	1	bit 6 cleared

Toggle a Bit

$x \wedge= (1 \ll 0) ;$ XOR flips bit 0

Run it twice and you are back where you started

	7	6	5	4	3	2	1	0	
x	0	0	0	0	1	0	0	1	
$1 \ll 0$	0	0	0	0	0	0	0	1	mask
$x \wedge \text{mask}$	0	0	0	0	1	0	0	0	bit 0 flipped

Test a Bit

Shift it down, then keep only bit 0:

`( x >> 3 ) & 1` 1 or 0

Or mask it in place:

`( x & ( 1u << 3 ) ) != 0` true or false

In Rust the second form gives a bool
ready to use in an if

The Precedence Trap

`if ( x & MASK == 0 )`

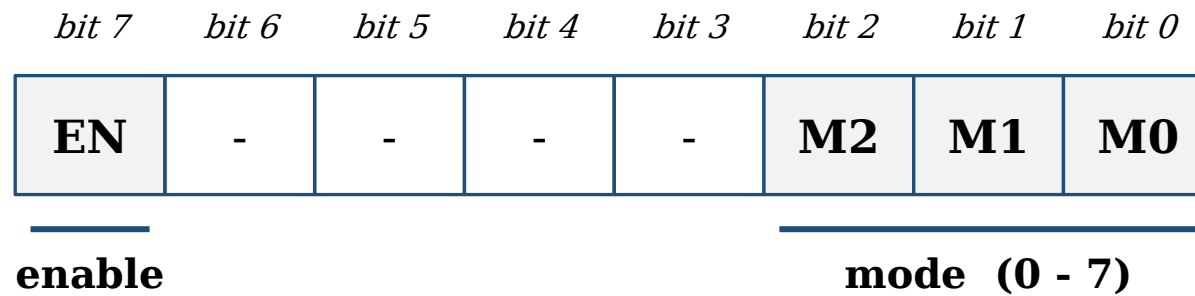
C, C++ read it as: `x & ( MASK == 0 )`
always 0 - the test never
passes

Rust reads it as: `( x & MASK ) == 0`

Always parenthesize bit operations

Fields Packed in a Register

One byte, two fields:



Reading and Writing Fields

```
mode = reg & 0x07 ;           low three bits  
enable = ( reg >> 7 ) & 1 ;   top bit
```

Change the mode, keep everything else:

```
reg = ( reg & ~0x07 ) | ( new_mode & 0x07 ) ;
```

Rust: the same, with `!` in place of `~`

Bit Operators on Structs

Members are integers - every bit operator applies:

```
my_sample.adc_counts & 0xFFF
```

keep only the 12 bits a 12-bit ADC

produces

```
my_sample.adc_counts >> 4
```

drop the 4 noisiest bits

Bit Operators on Arrays

A simple checksum: XOR every byte together

C, C++:

```
uint8_t check = 0 ;  
for ( int i = 0 ; i < N ; i++ ) check ^= data[i] ;
```

Rust:

```
let mut check : u8 = 0 ;  
for i in 0..N { check ^= data[i] ; }
```

Precedence in C and C++

Highest

! ~ ++ --

unary

* / %

+ -

<< >>

< <= > >=

== !=

& then ^ then |

&& then ||

= += -= &= |= ...

Lowest

Precedence in Rust

Highest

! -

as

* / %

+ -

<< >>

& then ^ then |

== != < <= > >=

&& then ||

= += -= &= |= ...

Lowest

unary

type conversion

cannot chain

What We Can Now Do

Arithmetic on variables, members, elements

Comparison values, and whole structs and
arrays

Clause logic && || ! on conditions

Bit logic & | ^ ~ << >> on every bit

Seven truth tables describe all of it

Let's practice

Exercise: Using **opspractice**